Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the Course: B.Tech (CSE)
Name of the Paper: Fundamental of IoT

Semester: 3rd
Paper Code: TCS331

Time: 3 Hours

Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	**(20marks)**	
(a)	1. Compare various identity management models in IoT—Local, Network, Federated, and Global Web Identity—highlighting their advantages and limitations. 2. Differentiate between user-centric, device-centric, and hybrid identity management approaches, giving practical use-case examples.	CO1, CO6
(b)	Explain how IoT-based object identification and mobility support can improve efficiency in precision agriculture. What environmental and traffic parameters should be considered?	CO2, CO5
(c)	A remote oil rig relies on satellite-based IoT devices for equipment monitoring: 1. Design a communication architecture that supports device mobility and minimizes power consumption. 2. Discuss how environmental conditions influence traffic characteristics and propose countermeasures. 3. Create a flowchart showing the system's data flow.	CO4
Q2	**(20 marks)**	
(a)	A city's energy department is deploying IoT-enabled smart meters to monitor consumption and forecast demand: 1. Recommend an identity management solution to ensure secure access and protect user privacy. 2. Discuss how IoT scalability principles can be applied to manage an increasing number of meters. 3. Assess the role of open architecture in enabling integration with third-party applications.	CO2, CO3 CO1,CO5 CO4
(b)	1. Explain the architecture of Wireless Sensor Networks (WSNs), their importance in IoT applications, and the challenges they encounter. 2. Draw a flowchart showing interactions among device intelligence, communication features, and mobility support in IoT systems.	
(c)	Evaluate the contribution of IoT in autonomous vehicles. Explain how communication capabilities and device intelligence enhance safety and performance, and suggest measures to mitigate potential risks.	
Q3	**(20 marks)**	
(a)	1. Differentiate between user-centric and hybrid identity management models in IoT. 2. Identify and discuss the major challenges in implementing clustering for large-scale IoT networks.	CO2
(b)	Develop a comprehensive IoT framework that addresses critical aspects of resource management, security, scalability, and interoperability by integrating advanced clustering techniques, software agents, identity management, and key IoT technologies.	CO4
(c)	1. Define clustering in the context of IoT and explain its significance. 2. Which clustering method can be used to improve resource utilization in a smart home IoT network?	CO3
Q4	**(20 marks)**	
(a)	In a smart city project utilizing Wireless Sensor Networks (WSNs) to monitor traffic,	CO1, CO6

	pollution, and energy usage: 1. Discuss the main challenges in configuring and securing such networks. 2. Explain how routing protocols can maintain efficient data transmission.	
(b)	1. Describe the structural components of IoT, emphasizing environment characteristics, traffic characteristics, and open architecture.	
	2.Explain why interoperability and open architecture are vital for IoT systems and how they enhance system performance.	CO2, CO5
(c)	1.Define "device intelligence" and "communication capabilities" in IoT. How do these features enable mobility, low power consumption, and seamless data exchange? 2.Explain how key IoT technologies—such as sensors, RFID, and satellites—help manage traffic parameters like latency, bandwidth, and packet loss.	CO2,CO3
Q5	**(20 marks)**	
(a)	1. Explore how IoT and WSNs work together in energy distribution systems to improve efficiency and reduce power loss. 2. Discuss the importance of power optimization strategies in battery-operated IoT devices, with examples.	CO5, CO6
(b)	Define Machine-to-Machine (M2M) communication in the context of the Internet of Things (IoT). Explain how M2M enables autonomous data exchange between devices without human intervention. Illustrate its role in enhancing industrial automation or smart city applications.	CO3, CO4
(c)	1. Design a simplified flow diagram showing how data flows between IoT sensors, M2M gateways, and cloud platforms in AgriSense's smart farming system. 2. Identify the communication protocols suitable for large-scale M2M connectivity in agricultural IoT systems. Discuss how they address energy efficiency and scalability.	CO1,CO2